

NOETHER: Constructive Metamorphic Pattern Identification from Operator Algebras and a Falsifiable Invariance-Blindness Theorem

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Context. Metamorphic Testing depends on metamorphic relations (MRs), yet MR identification remains largely experience-driven or search-driven. Expert MR sets often encode useful but implicit domain intuition, while generated candidates can be difficult to interpret, deduplicate, and maintain without an explanation of where the relation comes from and where it stops applying.

Objective. This paper studies MR identification rather than MR fault-revealing effectiveness. We ask whether a structured, operator-algebraic method can derive a broader and more explainable MR class / MetaPattern design space than expert MR sets and search-based MR generation, by making the algebraic origin and applicability boundary of MRs explicit from program-family governing equations.

Method. We introduce **NOETHER**, a two-level framework for equation-governed program families. The upstream layer curates a program-family operator algebra and decomposes it into recurrent operator blocks, including symmetry, order, self-adjointness, time reversal, limits, qualitative dynamics, method comparison, and relational equivalence. The downstream layer, CONSTRUCT-MP, derives MR classes / MetaPatterns from those blocks and then checks whether a specific program interface exposes the inputs and observations needed to instantiate an executable MR. The downstream derivation is provable: CONSTRUCT-MP is closed under the derivation operator, a well-formedness guarantee, and for the symmetry and self-adjoint blocks an *Invariance-Blindness Theorem*

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characterises exactly which implementation faults an algebra-derived MR can and cannot detect, converting that closure into a falsifiable, non-tautological prediction.

Evidence. We evaluate NOETHER as an MR-identification method along three axes. First, we compare expert MR sets and NOETHER using binary operator-block coverage. Second, we compare NOETHER with search-based MR generation on origin explanation, boundary explanation, redundancy, readability, and maintenance. Third, we trace cross-domain derivations showing that the same operator block can explain MR classes across different program families and solvers. Mutation and head-to-head results are retained only as secondary executability and sanity-check evidence.

Conclusion. NOETHER contributes a theory-guided MR-identification framework: it derives MR classes from governing-equation operator algebra, makes their origin and boundary auditable, and exposes structural MR design-space coverage that is not explicit in expert MR sets or search-generated candidates alone. A held-out evaluation confirms the Invariance-Blindness prediction, and a negative instance on a reactor-physics algebra bounds the method’s completeness and localises where the derivation must be extended.

CCS Concepts: • **Software and its engineering** → **Software testing and debugging**; *Software verification and validation*; • **Theory of computation** → *Algebraic semantics*.

Additional Key Words and Phrases: metamorphic testing, metamorphic relation identification, MetaPattern, operator algebra, program-family testability, test oracle problem, software testing foundations

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1 Introduction

Noether’s first theorem replaced an empirically curated catalogue of conservation laws with a derivation from the structure of an action functional.¹ A catalogue of observed invariants can sometimes be replaced by a derivation procedure grounded in the underlying structure. The replacement does not eliminate empirical work; it lifts the empirical step from per-instance enumeration to per-domain structural identification, one cycle of induction per stable domain rather than per program.

Software testing faces a related problem. Metamorphic Testing (MT), introduced by Chen et al. in 1998 [?], was admitted into the IEEE/ISO/IEC software-testing standards in 2022 [?] and has been increasingly endorsed as a technique for testing AI and machine-learning systems [? ?]. Its central artefact is the *metamorphic relation* (MR): a property that constrains how a program’s outputs must covary across multiple executions, thereby substituting for an explicit oracle in domains where one cannot exist. Despite MT’s maturity, MR identification, the task of deciding *which* properties hold for a program under test, remains its binding constraint. Practitioners report three recurring difficulties: high dependence on tester domain knowledge, mutually incompatible MR formulations for the same program, and low reuse of MR sets across teams or projects [? ?]. Recent work has responded at the application layer, by mining MRs for particular domains, and at the integration layer, by automating search through evolutionary, mining-based, or LLM-assisted pipelines [? ? ?]. The foundational layer has not advanced at the same pace.

The principal artefact at that foundational layer is the *MetaPattern* (MP): an equivalence class over MRs that captures a recurrent structural strategy a tester invokes when reasoning about program properties. MetaPatterns organise the

¹The theorem appeared in Noether’s 1918 paper *Invariante Variationsprobleme*, addressing a question Hilbert and Klein had raised about energy conservation in general relativity. We do not invoke it as a theorem about programs: program semantics and metamorphic relations do not provide an action functional. The analogy is methodological only.

otherwise unbounded MR design space into a small, interpretable scaffold; methods that exploit them, including the structured MR identification approaches METRIC [?] and METRIC+ [?], and the recent wave of LLM-prompted MR generators, depend on the scaffold’s quality. Yet across the literature, MetaPattern catalogues continue to be assembled in the same way conservation laws were assembled before 1918: by induction over observed examples. A typical proposal lists k patterns drawn from cluster analysis or expert codification, demonstrates that the patterns “cover” some corpus of MRs to a target threshold, and stops. None of the existing MP proposals, including the authors’ own prior work on five reactor-physics patterns, answers the three questions that any structural account of MetaPatterns should answer:

- (1) **Origin.** *Why* exactly these MetaPatterns and not others? What is the structural source of an MP, as distinct from an empirical regularity in the corpus on which it was induced?
- (2) **Closure.** Under what mathematical conditions is a discovered MP set *closed under a stated derivation operator*, in the sense that it is guaranteed not to miss patterns reachable through the operator from a structurally fixed input? Absolute completeness over all properties one might write over the underlying structure is a strictly stronger demand and remains open in general.
- (3) **Transferability.** When the program family changes (from reactor physics to natural-language inference, from numerical libraries to recommender systems), how does the MP set change, and can the new set be obtained without re-running the entire empirical induction in the new domain?

We call this the *origin–closure–transferability gap*. The gap matters because it explains why MR sets continue to grow as one-off artefacts. Without a structural source, there is no clear boundary on what the pattern space contains. Without a transfer rule, a relation written for one program rarely moves cleanly to another, even when both programs share a deep structural template. These are not only tooling limitations; they follow from grounding MetaPatterns in observed examples rather than in the structure that makes the relations hold.

This paper proposes such a structural source. We introduce **NOETHER**, a constructive framework that derives MetaPatterns from the operator-algebraic structure of the program family under test. The framework extracts MetaPatterns from invariants of an operator algebra $\mathcal{A}_P$ (formally defined in Section ??) that captures the program family’s mathematical scaffolding. Given $\mathcal{A}_P$, NOETHER produces a MetaPattern set $M(\mathcal{A}_P)$ (constructed by the algorithm of Section ??) together with a closure guarantee over the algebra-induced MR space. Given a different program family with a different algebra $\mathcal{A}_{P'}$, the same construction produces $M(\mathcal{A}_{P'})$ without re-running empirical induction. The framework does not abolish induction: domain experts must still distil $\mathcal{A}_P$ from program semantics, and Section ?? states this limitation explicitly. What becomes algebraic is the downstream step from $\mathcal{A}_P$ to the MetaPattern set. Figure ?? summarises the architecture.

We make five MR-identification contributions. The load-bearing theoretical result is an Invariance-Blindness Theorem: a falsifiable characterisation of which implementation faults an algebra-derived MR can and cannot detect, distinct from the intentionally modest by-construction closure guarantee.

- **C1.** We introduce NOETHER, a *layered* framework for MR-class / MetaPattern identification that combines an empirically curated structural decomposition (Hypothesis ??; upstream layer, itself partly distilled from the present authors’ prior PWR MR catalogue, see §?? for provenance) with a constructive algorithm for deriving MetaPatterns from a program-induced operator algebra (downstream layer).
- **C2a (positive theory).** Given the structural decomposition, Theorem 1 proves a *no-drop closure invariant*: every MR that is reachable by the framework’s Translate operator over the algebra-induced MR space $MR(\mathcal{A}_P)$ is

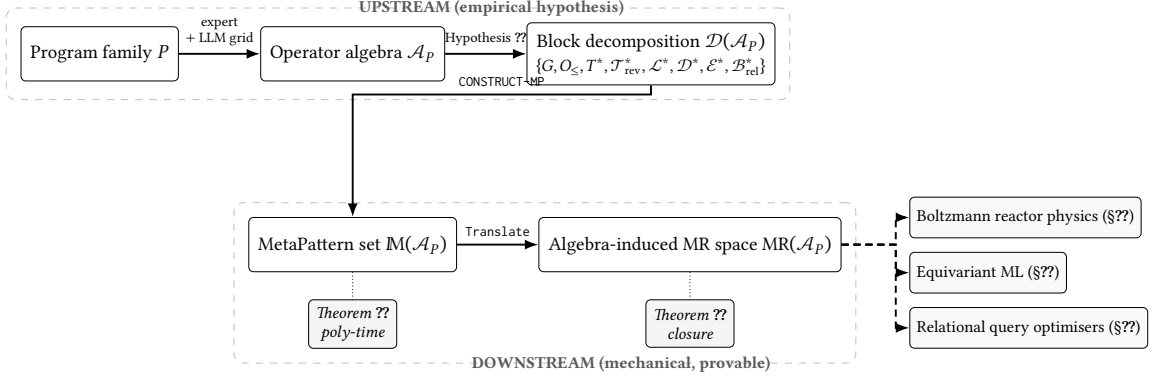


Fig. 1. NOETHER framework architecture. The *upstream layer* curates the program family’s operator algebra $\mathcal{A}_P$ and its structural decomposition as an empirical hypothesis (Hypothesis ??). The *downstream layer* mechanically derives the MetaPattern set $M(\mathcal{A}_P)$ via CONSTRUCT-MP, with closure under Translate over $MR(\mathcal{A}_P)$ (Theorem ??) and polynomial-time constructibility under a finite generating set (Theorem ??). The framework is instantiated on three structurally distinct domains (§??, §??, §??) to test transferability at the algebra-skeleton level.

assigned to a canonical operator block and MetaPattern class. This result is intentionally modest. It is a well-formedness guarantee for the downstream construction, not evidence that the upstream block list is complete and not evidence of fault-detection superiority. Section ?? states the explicit scope of Definition ??, and Remark ?? documents the MR classes outside that scope. Polynomial-time constructibility of CONSTRUCT-MP holds when the algebra admits a finite generating set (Theorem 2).

- **C2b (negative theory).** The strictly stronger absolute-completeness conjecture (Theorem 1’ / Conjecture ??), over arbitrary properties expressible in $\mathcal{A}_P$, is *false* on $\mathcal{A}_{PWR}$: two MRs from the standard PWR safety-analysis literature (non-additivity of rod-bank reactivity worth; second-order mixed dependence of k_{eff} on moderator temperature and boron concentration) are formulable over $\mathcal{A}_{PWR}$ but not in $MR(\mathcal{A}_{PWR})$ (§??, Appendix ??). The two MRs identify five structural obstructions in Translate’s signature, with pairwise independence established by per-block exhaustion. Companion surveys on $\mathcal{A}_{equi}$ and $\mathcal{A}_{rel}$ (§??) identify five further candidate dimensions (two on $\mathcal{A}_{equi}$ specialising PWR-side dimensions; three net on $\mathcal{A}_{rel}$), yielding five Translate-extension dimensions proved pairwise-independent on $\mathcal{A}_{PWR}$ plus five candidate dimensions on $\mathcal{A}_{equi}$ and $\mathcal{A}_{rel}$ whose pairwise independence is asserted by inspection; formal per-dimension exhaustion on those algebras is committed as follow-up. Scope statements are in Section ??.
- **C2c (limiting theory).** We prove the *Invariance-Blindness Theorem* (§??): for the symmetry (G) and self-adjoint (T^*) blocks, an algebra-derived MR’s detection kernel equals *exactly* the structure-preserving faults, within the linear operator-implementation fault class and under a faithfulness condition that a finite test attains. Corollaries: a single-block battery is structurally incomplete; completeness requires oracle families with trivial joint kernel; a neutral cross-implementation differential oracle is the complementary (common-mode) oracle. This converts the by-construction closure of C2a into a falsifiable, non-tautological characterization of the derivation’s blind spots; tightness beyond $\{G, T^*\}$ and beyond the linear fault class is delimited in §??.
- **C3.** We instantiate NOETHER on a real-world program family and show how it *systematises and re-classifies* previously catalogued patterns. The framework re-classifies the prior inductive catalogue under a uniform algebraic

structure: it reproduces three prior MetaPatterns, refines two on a sounder algebraic basis, and de-duplicates two equivalence classes (m_{adj} , m_{rev}) that the inductive catalogue had not isolated as structurally distinct. These are not de novo discoveries; NOETHER’s contribution is the algebraic warrant for treating them as separate MetaPattern slots within a canonical block ordering, and the deflationary direction (revealing over-counting on some sub-families) constitutes a further non-circular form of re-classification. The deflationary direction is non-circular relative to the prediction caveat of §??: §?? demonstrates it on three independent test cases (a sorting library; Murphy et al. 2008’s categorisation of MRs for ML into six classes [?] re-decoded against a feedforward classifier algebra; the prior reactor-physics catalogue itself), without invoking T^* or $\mathcal{T}_{\text{rev}}^*$ structure that was curated from reactor physics.

- **C4.** We demonstrate *structural* transferability at the algebra-skeleton level (not cross-domain empirical superiority), within the framework’s scope precondition, by instantiating NOETHER on three structurally distinct operator-algebraic domains at differentiated evaluation depths: Boltzmann reactor-physics transport (Section ??), equivariant ML (Section ??), and relational query optimisers (Section ??); the last exercises the relational-equivalence block, whose algebraic skeleton differs from the Lie-group / self-adjoint / time-reversal core.
- **C5.** We define an evidence protocol for MR identification: binary expert-vs-NOETHER operator-block coverage, search-vs-NOETHER origin/boundary comparison, and compact cross-domain derivation traces. Existing mutation, GenMorph, and DeepCrime-style results are treated as secondary executability checks, not as evidence of average fault-detection superiority.

Scope of contribution. This paper evaluates *MR identification*: whether a method can systematically derive structurally justified MR classes from a program family’s governing equations. It does not evaluate MR *effectiveness* in the sense of average fault-detection rate, mutation score, or general defect-revealing superiority. The eight blocks are curated by inspecting mathematical structures that recur across the program families we have studied; they are not derived from an algebraic axiom. The framework therefore has two layers: an *upstream layer* (curating $\mathcal{A}_P$ and its block decomposition) that remains empirical and human, and a *downstream layer* (mechanically deriving $M(\mathcal{A}_P)$ from $\mathcal{A}_P$) that is algorithmic and provable. We do not claim to have eliminated induction from MetaPattern discovery. We move induction one level up, from “what MetaPatterns recur in observed MR samples?” to “what algebraic structures recur in the program families practitioners care about?”, and make the downstream step mechanical.

Boundary of contribution

This paper establishes:

- (1) A no-drop closure invariant for the downstream construction: given a block decomposition, $M(\mathcal{A}_P)$ is closed under `Translate` over the algebra-induced MR space $\text{MR}(\mathcal{A}_P)$ for the stated operator algebras (Theorem ??; by-construction within the explicit scope of Definition ??, see Remark ??). Closure over the strictly larger space of arbitrary properties expressible in $\mathcal{A}_P$ is Theorem 1’, falsified on $\mathcal{A}_{\text{PWR}}$ (see item (a) below);
- (2) Polynomial-time constructibility of CONSTRUCT-MP under explicit complexity assumptions (Theorem ??);
- (3) Three non-vacuous instantiations: Boltzmann reactor physics (§??), equivariant ML (§??), and relational query optimisers (§??); the relational-equivalence block $\mathcal{B}_{\text{rel}}^*$ extends the framework beyond the Lie-group / self-adjoint / time-reversal mathematical core;
- (4) The Invariance-Blindness Theorem for the G and T^* blocks (Theorem ??): the symmetry / self-adjoint MR detection kernel equals the structure-preserving faults (faithfulness-tight), within the linear operator-implementation fault class.

It does *not* establish:

- (a) Absolute completeness over arbitrary properties expressible in $\mathcal{A}_P$. This is Theorem 1' (Conjecture ??, Appendix ??); §?? establishes that it is false on the PWR core diffusion algebra $\mathcal{A}_{\text{PWR}}$ via two independent counterexamples (non-additivity of rod-bank reactivity worth, and second-order mixed dependence of k_{eff} on moderator temperature and boron concentration), identifying five structural obstructions in Translate's signature whose pairwise independence is proved by per-block exhaustion (Appendix ??). Companion surveys on $\mathcal{A}_{\text{equi}}$ and $\mathcal{A}_{\text{rel}}$ (§??, companion artefacts in theory/) identify five further candidate dimensions on those algebras (so the falsification is not PWR-specific). Pairwise independence on the candidate five is asserted by inspection rather than by formal exhaustion proof; full per-dimension exhaustion proofs on $\mathcal{A}_{\text{equi}}$ and $\mathcal{A}_{\text{rel}}$ are committed as follow-up. The question of whether a Composite-Translate extension absorbs the ten Translate-extension dimensions identified across the three algebras while preserving Theorem ??'s closure and Theorem ??'s polynomial-time constructibility remains open.
- (b) Sufficiency of the structural list. Hypothesis ?? is an empirical hypothesis with six enumerated out-of-scope program-family classes (Remark ??); each such class is a candidate ninth block.
- (c) Superiority over existing automated MR-identification pipelines on average. The comparative material in §?? is retained as secondary executability and sanity-check evidence; it does not characterise the framework's behaviour on arbitrary defect distributions.
- (d) Elimination of induction from MetaPattern discovery. Induction is *relocated* from MR-instance level to algebra-block level, not eliminated.

The remainder of the paper follows a TOSEM-style method-paper structure. Section ?? reviews expert MR design, structured taxonomies, and search-based MR generation. Section ?? presents the NOETHER method, the two-level model from identified MR class to executable MR instance, the operator-block taxonomy, and the coverage rules. Section ?? defines the three evaluation questions and evidence protocol. Section ?? reports expert-set coverage, search-method complementarity, and cross-domain derivation evidence. Section ?? discusses threats to validity and limitations. Section ?? states future work; Section ?? concludes.

2 Related Work

We organise prior work along four lines: MT/MR fundamentals and the long-standing identification bottleneck; structured MR identification through category, specification, and model scaffolds; automated MR identification through mining, search, learning, and LLM assistance; and MetaPattern catalogues, including our prior reactor-physics taxonomy. These lines differ in mechanism, but they share one limitation: their pattern structures are induced from observed MRs, mined artefacts, or manually supplied models rather than derived from the mathematical structure that makes those MRs hold.

2.1 Metamorphic testing and the MR identification bottleneck

Metamorphic Testing was introduced by Chen, Cheung, and Yiu in 1998 to address the test-oracle problem [?], a general obstacle in software testing when expected outputs cannot be automatically determined [?]. An MR is a logical implication of the form $R_i(x_1, \dots, x_n) \Rightarrow R_o(P(x_1), \dots, P(x_n))$, where P is the program under test, R_i is an input relation, and R_o is an output relation. When R_i holds but R_o fails on actual executions, a fault is reported, without any need to know the absolute correct output of any single execution. Over two decades MT has become standard equipment for testing systems whose oracles are otherwise inaccessible: scientific computing, machine learning classifiers, autonomous vehicles, search engines, compilers, and large language models [? ?].

The community has long acknowledged a single binding constraint on MT’s effectiveness: MR identification. Surveys spanning twenty years agree that identifying high-quality MRs requires (i) deep familiarity with the program’s functional semantics, (ii) substantive domain background (physical, mathematical, or linguistic, depending on the system), and (iii) the ability to convert that background into executable property assertions [? ?]. Liu et al. [?] provide the canonical empirical evidence that even ad-hoc MRs detect a substantial fraction of mutants given modest tester training, while Murphy et al. [?] catalogued the structural properties of ML applications relevant to MR design and Xie et al. [?] demonstrated that MT scales effectively to supervised classifiers. Testers without sufficient domain background tend to identify only “trivial MRs”. A 2024 survey explicitly identifies AI assistance, especially via large language models, as the most promising open avenue for closing the identification gap [?]; Saha and Kanewala [?] report that, on a benchmark of 709 mutants for supervised classifiers, current MR sets detect only 14.8%, which exposes the gap.

Beyond difficulty, the MR identification bottleneck has a more troubling structural feature: even when MRs are identified, the field has accumulated little consensus on *how* they are identified. Different authors confront the same program and produce dissimilar MR sets. Once written, an MR rarely migrates: a relation drafted for one ML classifier seldom transports to another, and a relation drafted for one numerical solver seldom transports across solver families.

2.2 Structured MR identification

Structured MR identification has already moved the field away from ad hoc relation design. METRIC organises MR construction around an “input/output category” framework: the tester first identifies relevant categories of input transformations and output relations, then composes MRs from category pairs [?]. METRIC+ extends this scheme by enriching the category catalogue and providing systematic combination rules that reduce the human burden in category enumeration [?]. Other structured lines instantiate the scaffold through domain or artefact models: Segura et al. define MR patterns for query-based systems from recurrent query features and explicitly use relational algebra as the vocabulary for those patterns [?]; Li et al. construct MRs from tabular-expression specifications [?]; Yan et al. decompose the input side of scientific-computing MRs into basic input-pattern structures [?]; and Lin et al. organise scientific-software MRs hierarchically [?]. Together, these approaches remain the strongest existing attempt to give MR identification an explicit scaffold.

Our disagreement is not with this direction but with its grounding. The categories, patterns, tables, and input decompositions are introduced through expert curation, artefact-specific structure, or observed reuse, and are validated by empirical coverage on benchmark programs. This leaves two of our three foundational questions unanswered. *Origin*: existing structured methods do not derive their categories from program-level operator algebra. Even query-based MR patterns use relational algebra as an organising vocabulary, not as a closed derivation system whose primitive operator blocks define an MR design space. *Closure*: these frameworks do not provide a mathematical condition under which the category set is complete; coverage is reported but not proved. The third question, *transferability*, is only partially addressed. The same category templates can be invoked across domains, but because the categories are not algebraically bound to specific program structures, transfer rests on an assumption of universality that remains open.

2.3 Automated MR identification

Automated MR identification attacks the bottleneck by searching for executable relations more directly. Early search and dynamic-inference work includes likely metamorphic-property inference from executions [?], search-based inference of polynomial MRs [?], and AutoMR’s discovery and cleansing of numerical MRs through search plus redundancy reduction [?]. MR-Scout mines MRs from existing test suites by extracting input-transformation and output-assertion

patterns from test-case pairs and abstracting them into reusable relations [?]. GenMorph evolves MR candidates through genetic programming, co-evolving input transformations with output assertions and using mutation-killing as the fitness signal [?]. Shin et al. derive executable MRs from natural-language requirements through few-shot prompting of a large language model, with validation through an industrial questionnaire study with Siemens [?]; further LLM-assisted variants, including ChatGPT-driven MR generation [?], domain-customised GPTs for autonomous-driving simulators [?], and multi-agent retrieval-augmented pipelines for traffic-rule MRs [?], extend this template into safety-critical or rule-rich domains. Within ML-system testing, differential-testing approaches such as DeepXplore [?] use input-transformation invariance as a proxy oracle, complementary to the explicit MR formulation we adopt here.

A second strand uses program-structure, specification, documentation, or model features rather than test-case mining. Kanewala et al. [?] predict MRs of scientific software via graph-kernel learning over control-flow and data-dependency graphs, an early demonstration that program structure carries enough signal for automated MR discovery. MeMo extracts equivalence MRs from Javadoc comments [?]. Nolasco et al.’s MemoRIA [?] infers MRs by deriving an object-protocol abstraction and validating candidates through fuzzing plus SAT-based reduction, then evaluates on 22 Java subjects. Clark et al. generate MRs from causal graphs, making model misspecification an explicit boundary of the method [?]. Tao et al.’s Mettoc [?] applied MT to compiler testing using equivalence-preservation as the canonical relation. Ying et al. [?] formalise relationships between metamorphic-relation patterns into family trees, and Altamimi et al. [?] provide a recent systematic literature review of MR-automation work that catalogues the structural-vs-empirical-search divide our framework targets. Earlier LLM-assisted attempts include Zhang et al. [?] on ChatGPT-driven MR generation.

Each of these methods advances automation on a different axis, but all treat the MR space as an empirical, artefact-derived, or model-supplied search space. MR-Scout’s recall is bounded by the relations already latent in existing tests. Search-based and evolutionary methods can discover useful numerical or executable MRs, but the reported cleansing and redundancy-reduction steps are consequences of the search representation rather than explanations of the mathematical origin of the surviving relations. GenMorph’s evolutionary search is shaped by the fitness landscape induced by mutation killing. LLM-prompted approaches are sensitive to prompt phrasing when no structural prior constrains the generation. Documentation-, protocol-, and causal-model-based methods are more interpretable, but their boundary is the supplied artefact or model rather than a program-family operator algebra. None of these methods can state, before search begins, which MR types lie outside its reach, because none has an algebraic account of the space being searched.

The intuition that symmetries furnish partial oracles predates this work. Gotlieb’s symmetric testing treats input permutations whose outputs must agree as a checkable oracle [?], and a follow-up study models its test quality on Java Card APIs [?]. Patel and Hierons survey such oracle strategies for non-testable systems [?], while Khritankov and Iakusheva systematically review metamorphic-relation derivation and group the recurring input transformations into six families [?]. In machine learning, related symmetry notions are studied empirically: Gruver et al. measure how much equivariance a trained network actually acquires [?], and Kaba et al. attain equivariance by learning canonicalization functions rather than constraining the architecture [?]. These lines name and exploit the symmetry-oracle intuition, yet each identifies relations per program, surveys them descriptively, or approximates symmetry inside a model. None derive metamorphic patterns algorithmically from the governing equations of a program family. NOETHER does not invent the symmetry layer; its contribution is a constructive derivation over the operator-block layer, together with

closure and complexity results and a domain-transferable skeleton, turning the symmetry oracle from a per-program insight into a provable construction.

2.4 MetaPattern catalogues and empirical adequacy

MetaPattern catalogues organise the post-hoc inventory of identified MRs into a smaller vocabulary of recurring strategies. Such catalogues typically result from clustering observed MRs, codifying expert intuitions about “kinds of properties testers reason about”, and validating the catalogue’s coverage on a benchmark MR corpus. Reactor-physics test suites have produced taxonomies of conservation, monotonicity, convergence, trajectory, and partial-order patterns [? ?]; Zhou et al. [?] introduced an explicit *symmetry metamorphic relation pattern* as a reusable abstraction for deriving concrete MRs across applications; Lin et al. [?] proposed hierarchical MRs for scientific software; Ying et al. [?] formalises the relationships between MR patterns into *family trees* with explicit refinement and specialisation edges, and reports a SOTA pattern hierarchy that organises previously proposed patterns into a single classification structure. The most recent state-of-the-art survey of MR generation [?] maps the field’s twenty-year output across pattern catalogues, mining, evolutionary search, and LLM-prompted methods. Adequacy frameworks such as the Pattern–Matrix Coverage Metric, and recent MT adequacy work based on MRs and source inputs [?], assess how thoroughly a given test suite or MR set occupies an already-defined testing space. They do not define coverage over the operator blocks that give rise to MR classes.

Ying et al.’s family-tree formalism [?] is the closest published cousin to NOETHER’s MetaPattern equivalence-class structure: both organise MR patterns into a hierarchy with explicit relationships between patterns. The two formalisms differ in the relationship type. Ying et al.’s family tree is a *refinement/specialisation tree* rooted in informally-named pattern categories (symmetry, additive, multiplicative, etc.); NOETHER’s MetaPattern equivalence classes are *quotients of the algebra-induced MR space* $\text{MR}(\mathcal{A}_P)$ under structural equivalence (Definition ??), with each class derived from a specific block of $\mathcal{D}(\mathcal{A}_P)$. A family-tree node in Ying et al. typically corresponds to one or more NOETHER MetaPatterns when the node admits an operator-algebraic specification (e.g. Ying et al.’s “symmetry” parent node decomposes into the G -block MetaPattern m_{inv} for finite-group symmetries plus the T^* -block MetaPattern m_{adj} for self-adjoint dualities under NOETHER’s structural decomposition); conversely, NOETHER’s $\mathcal{B}_{\text{rel}}^*$ MetaPattern has no direct counterpart in the family-tree formalism because relational-algebra equivalences were not in Ying et al.’s benchmark set. The two formalisms are complementary: Ying et al. catalogues patterns inductively at the MR-instance level; NOETHER provides an algebraic warrant for the existence of each pattern, given the operator algebra. A unified family-tree-plus-algebra-block reading of the literature would map each family-tree node to (at most) one NOETHER block when the node has an algebraic source, and would flag family-tree nodes without algebraic source as candidates for the ninth-block list of Remark ??.

Three further pattern-catalogue or MR-grammar candidates raised in peer review (Hu et al. 2019; Mariani 2018; Liu et al. 2020) could not be located through the standard fallback chain (CrossRef, OpenAlex, DBLP, Semantic Scholar, Google Scholar) under the citation profiles provided, and are therefore not cited; the closest verifiable cousins in this space are Zhou et al. [?], Lin et al. [?], and Ying et al. [?].

Such catalogues are useful, but each is, by construction, an empirical artefact. None of the existing catalogues, including the present authors’ own, can answer the three foundational questions of Section ??.

2.5 Convergent diagnosis

Across all four lines of work, two problems recur. *Unbounded MR emergence*: every new domain or new program produces fresh MR formulations. *Poor MR reusability*: identified MRs do not transport readily across programs that share deep structural similarities. Both problems follow from grounding MetaPatterns inductively. NOETHER replaces that inductive grounding with operator-algebraic grounding.

Relation to a separate line of work on minimum complete MR subsets. A separate line of work by some of the present authors addresses the *orthogonal* problem of *selecting* a minimum complete subset from a *given* metamorphic-relation pool: taking an MR set as input, it characterises the smallest subset that preserves the pool’s fault-detection power under a fixed fault model, proves the corresponding decision problem NP-hard, and provides approximation and exact (ILP) solvers. That work takes the MR set as given and optimises its cardinality; the present paper is concerned with the upstream question of where MRs and MetaPatterns *come from*: their constructive derivation from a program-induced operator algebra and algebraic closure under `Translate`. The two are complementary: NOETHER *generates* the algebra-induced MR space that a subset-selection procedure would then *minimise*. No theorem or empirical claim is shared: the present paper makes no minimality or subset-selection claim, and the subset-selection work makes no derivation or closure claim.

Comparators in the head-to-head: what is compared and why. The empirical comparison in §?? runs Set N against a single executable baseline: Set G (GenMorph’s GP-evolved MRs at the published 30-min GAssert budget, [?]). The other three SOTA categories enumerated in the related-work survey contribute differently: METRIC+ [?] is contrasted as a category-enumeration scaffold (§??, §??) rather than as an executable fault-detection pipeline, because no public METRIC+ implementation auto-generates MRs from a $\mathcal{A}_P$ specification: a full PIT-based METRIC+ vs Set N comparison is committed as supplementary S4 (future_work.md) item (i); MR-Scout [?] is omitted from the executable head-to-head because its mining input (a pre-existing test corpus) is structurally absent on the algebra-rich Java substrate of §?? (the SUTs are stand-alone mathematical methods with synthesised test inputs, not codebases with developer-written tests latent in test files); AutoMT [?] and GPTMR [?] target safety-critical / traffic-rule autonomous driving domains rather than the operator-algebraic substrate, so an in-scope comparator selection from the LLM-assisted line uses an unprompted-by-NOETHER LLM ensemble (Set L; §??, §??) rather than these domain-specialised pipelines. Set B is a literature-MR baseline drawn from MT-for-ML references [? ? ?], restricted to point-cloud-classifier-applicable MRs. The single executable head-to-head against Set G is therefore the GP-evolved-baseline arm of a three-SOTA-category protocol; the LLM-assisted and mining-based arms are reported with their proper comparator (Set L ensemble at 2 vendors $\times$ 5 temperatures, §??, and the MR-Scout structural-absence rationale above), so the protocol’s categorical framing is preserved at the price of using different arms in different sections.

3 Proposed Method

This section presents NOETHER as an MR-identification method. The method takes a program-family governing equation/operator system as input, distils a program-induced operator algebra $\mathcal{A}_P$, decomposes the algebra into recurrent operator blocks, derives MR classes / MetaPatterns from block invariants, and finally checks whether a concrete program interface exposes the controls and observations needed to instantiate an executable MR. The primary output is an *identified MR class*; an executable MR instance is a downstream realization that may be blocked by missing interface affordances.

Counting rule used throughout the paper. Expert coverage is binary: an expert MR set covers an operator block if at least one expert MR belongs to that block. NOETHER coverage is also binary: NOETHER covers a block if the method explicitly derives at least one MR class from the program-family operator algebra in that block. Equation structure alone is not counted as coverage unless a corresponding MR class is written down. This rule is used before any empirical result is interpreted.

Operator-algebraic preliminaries and the structural decomposition appear in §??; the algebra-induced metamorphic relation space, the CONSTRUCT-MP construction, the algebraic closure theorem (Theorem ??), and the polynomial-time constructibility theorem (Theorem ??) in §??. The framework is then instantiated on three structurally distinct operator algebras: Boltzmann reactor physics in §??, equivariant machine learning in §??, and relational query optimisers in §??. The section concludes by falsifying the strictly stronger absolute-completeness conjecture, Theorem 1', on a fourth program family (PWR core diffusion) via two pairwise-independent counterexamples (§??).

3.1 Operator-algebraic preliminaries

This section introduces the algebraic apparatus used by NOETHER. We assume basic familiarity with group actions, equivalence classes, and quotient sets, but not with functional analysis. Each construct is defined at the level needed for the framework and later instantiated in reactor physics and equivariant machine learning.

3.1.1 Programs and program-induced operator algebras. Throughout the paper we treat a program P as a (possibly partial) computable function $P : \mathcal{X} \rightarrow \mathcal{Y}$. The class of programs of interest belongs to a *program family* $\mathcal{F}$ whose members share an underlying mathematical scaffolding.

DEFINITION 1 (PROGRAM-INDUCED OPERATOR ALGEBRA). *Let P belong to a program family $\mathcal{F}$. A program-induced operator algebra of $\mathcal{F}$ is a tuple*

$$\mathcal{A}_{\mathcal{F}} = (\mathcal{O}, \circ, \sim_{\mathcal{F}}),$$

where $\mathcal{O}$ is a set of operators acting on $\mathcal{X}$, $\mathcal{Y}$, or both; $\circ$ is an operator composition; and $\sim_{\mathcal{F}}$ is an equivalence relation declaring two operator expressions equal whenever they agree on every program of $\mathcal{F}$.

3.1.2 Symmetry groups (building block B1).

DEFINITION 2 (SYMMETRY GROUP OF $\mathcal{A}_P$). *A symmetry group of $\mathcal{A}_P$ is a subgroup $G \leq \mathcal{A}_P$ whose elements act on $\mathcal{X}$ such that, for all $P \in \mathcal{F}$ and all $g \in G$,*

$$P(g \cdot x) = \rho(g) \cdot P(x) \quad \forall x \in \mathcal{X},$$

where $\rho : G \rightarrow \text{End}(\mathcal{Y})$ is a (possibly trivial) representation of G on $\mathcal{Y}$.

3.1.3 Order operators: monotonicity and linearity (building block B2).

DEFINITION 3 (MONOTONE OPERATOR). *P is monotone with respect to θ if $\theta_1 \leq \theta_2 \Rightarrow P(\theta_1) \leq_{\mathcal{Y}} P(\theta_2)$ (or the reversed inequality, in which case P is anti-monotone).*

DEFINITION 4 (LINEAR OPERATOR). *P is linear on $\mathcal{X}_0 \subseteq \mathcal{X}$ when $P(\alpha x_1 + \beta x_2) = \alpha P(x_1) + \beta P(x_2)$ for all $x_1, x_2 \in \mathcal{X}_0$ and scalars α, β .*

3.1.4 Self-adjoint operators (building block B3).

DEFINITION 5 (SELF-ADJOINT OPERATOR). *Given an inner product $\langle \cdot, \cdot \rangle$ on $\mathcal{X}$, an operator $L \in \mathcal{O}$ is self-adjoint if $\langle Lx, y \rangle = \langle x, Ly \rangle$ for all x, y in the domain of L .*

Self-adjointness encodes a duality between the two arguments of an inner product. Reciprocity theorems in physics, transposed-graph identities in algorithms, and detailed-balance conditions in stochastic processes are all instances. In reactor physics, the adjoint transport formulation is self-adjoint under the appropriate inner product, yielding source-detector reciprocity.

3.1.5 Time-reversal operators (building block B4).

DEFINITION 6 (TIME-REVERSAL OPERATOR). *When $\mathcal{X}$ admits a time coordinate, a time-reversal operator $\mathcal{T}$ acts on inputs by reversing the time variable, $\mathcal{T}(x(t)) = x(-t)$. A program P is time-reversal symmetric on a sub-family of inputs when $P(\mathcal{T}x)$ is determined by $P(x)$ through a fixed bijection on $\mathcal{Y}$.*

Time-reversal applies only where the underlying dynamics admit a reversible sub-family. Dissipative dynamics break this symmetry, so the corresponding MetaPattern is empty for those systems.

3.1.6 Limit operators (building block B5).

DEFINITION 7 (LIMIT OPERATOR). *A parametrised limit operator $\mathcal{L}_\theta$ is a family of operators indexed by a parameter θ such that there exists a limit element $\mathcal{L}_*$ with $\mathcal{L}_\theta \rightarrow \mathcal{L}_*$ as $\theta \rightarrow \theta_*$ in an appropriate operator topology.*

3.1.7 Qualitative-dynamics operators (building block B6).

DEFINITION 8 (QUALITATIVE-DYNAMICS OPERATOR). *A qualitative-dynamics operator $\mathcal{D}$ is an operator on solution trajectories of an underlying ODE/PDE that extracts qualitative features such as extrema, inflection points, monotonic phases, overshoot magnitudes, S-curve transitions, and phase-portrait orbits, all of which are invariant under perturbations preserving the underlying dynamical structure (Sturm-type comparison theorems, dynamical-systems classification).*

Some MR-relevant invariants are not point-wise but *shape-wise*: a solution curve may have an overshoot, a single extremum, or a monotone-then-saturating profile. B6 gives these qualitative-dynamics MRs their own algebraic root.

3.1.8 Method-comparison operators (building block B7).

DEFINITION 9 (METHOD-COMPARISON OPERATOR). *A method-comparison operator $\mathcal{E}$ encodes a partial order $\preceq_{\mathcal{E}}$ on numerical or algorithmic methods, where $M_1 \preceq_{\mathcal{E}} M_2$ asserts that method M_1 produces an approximation no worse than method M_2 in a specified error norm, under specified conditions.*

3.1.9 *Decomposition of an operator algebra.* Given $\mathcal{A}_P = (\mathcal{O}, \circ, \sim_{\mathcal{F}})$, we decompose $\mathcal{O}$ along the eight building blocks introduced above (summarised in Figure ??):

$$\mathcal{D}(\mathcal{A}_P) = (G, O_{\leq}, T^*, \mathcal{T}_{\text{rev}}^*, \mathcal{L}^*, \mathcal{D}^*, \mathcal{E}^*, \mathcal{B}_{\text{rel}}^*),$$

where G collects symmetry subgroups, $O_{\leq}$ monotone and linear operators, T^* self-adjoint operators, $\mathcal{T}_{\text{rev}}^*$ time-reversal operators, $\mathcal{L}^*$ limit operators, $\mathcal{D}^*$ qualitative-dynamics operators, $\mathcal{E}^*$ method-comparison operators, and $\mathcal{B}_{\text{rel}}^*$ relational-equivalence operators.

Conservation laws are not a ninth block: by the Noether-style correspondence between symmetries and conserved quantities, a conservation/invariance relation is the G -block MetaPattern m_{inv} (Table ??), so where the EQ1 and EQ3

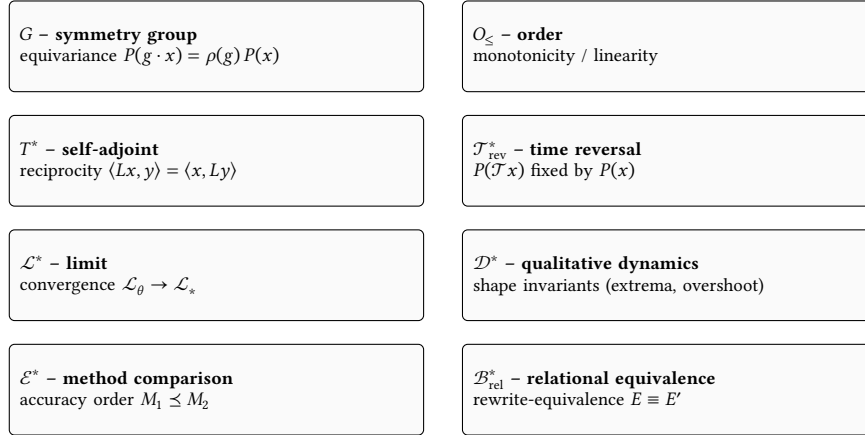


Fig. 2. The eight operator blocks of the decomposition $\mathcal{D}(\mathcal{A}_P)$, each shown with the structural invariant that gives rise to its MR class. The symmetry (G), self-adjoint (T^*), and time-reversal ($\mathcal{T}_{\text{rev}}^*$) blocks form the Lie-group / self-adjoint / time-reversal mathematical core; the Invariance-Blindness Theorem (Theorem ??) characterises the detection boundary of the G and T^* blocks, and $\mathcal{B}_{\text{rel}}^*$ extends the framework beyond the core. Blocks are populated per program family and may be empty (for example $\mathcal{T}_{\text{rev}}^*$ for dissipative dynamics, $\mathcal{B}_{\text{rel}}^*$ for families without idempotent-semiring rewriting structure).

coverage tables and the held-out transferability paragraph track a separate “conservation invariant” entry it denotes this conserved-quantity instance of the G block rather than a distinct operator block.

DEFINITION 10 (RELATIONAL-EQUIVALENCE BLOCK). A relational-equivalence operator is a binary relation $\equiv_{\mathcal{R}}$ on expressions over an idempotent semiring $(\mathcal{S}, \oplus, \otimes, \mathbf{0}, \mathbf{1})$ such that $E \equiv_{\mathcal{R}} E'$ iff E and E' are equal under all valid evaluation contexts of $\mathcal{S}$, generated by a finite set $\mathcal{R}_{\text{rel}} = \{\mathcal{R}_1, \dots, \mathcal{R}_K\}$ of identity-preserving rewriting rules, each $\mathcal{R}_i$ an ordered pair $(\text{lhs}_i, \text{rhs}_i)$ of semiring expressions with the property $\text{eval}(\text{lhs}_i, D) =_{\text{bag}} \text{eval}(\text{rhs}_i, D)$ on every valid input D . The block $\mathcal{B}_{\text{rel}}^*$ collects all such operators. It is empty for program families without idempotent-semiring rewriting structure (Boltzmann reactor physics, equivariant ML) and non-empty for those with it (relational query optimisers, §??).

Necessity, sufficiency, and the empirical status of the decomposition. We do not claim that the eight blocks are necessary in an absolute sense, that they exhaust all algebraic structures relevant to software testing, or that they follow from first principles. They are an *empirical curation*: a by-inspection enumeration of mathematical structures that recur across the program families we have studied. The claim is therefore scoped: the eight blocks are currently sufficient for these families, not provably necessary in general. This upstream empirical status is NOETHER’s main limitation as a theoretical framework. Induction has not been eliminated from MetaPattern discovery; it has been moved one level up, from “what MetaPatterns recur in observed MR samples?” to “what algebraic structures recur across program families?” Given such a curated decomposition, the downstream derivation of $\mathbb{M}(\mathcal{A}_P)$ becomes mechanical and provably closed in the sense of Theorem ???. We state the structural sufficiency as an explicit empirical hypothesis with a documented out-of-scope catalogue.

HYPOTHESIS 1 (DECOMPOSITION SUFFICIENCY). Let P be a program belonging to a program family $\mathcal{F}$ whose semantics admit an operator-algebraic representation $\mathcal{A}_P$ over (i) finite or finite-dimensional symmetry groups, (ii) partial orders with monotone or linear operators, (iii) self-adjoint operators with respect to a fixed inner product, (iv) anti-unitary involutions of a time coordinate, (v) limit operations of analytical (mesh-refinement, parameter-perturbation, or asymptotic) type,

(vi) qualitative-dynamics attractors of an underlying ODE/PDE, (vii) inter-method comparison partial orders, and (viii) identity-preserving rewriting rule sets on an idempotent semiring. For such families the structural decomposition

$$\mathcal{D}(\mathcal{A}_P) = G \dot{\cup} O_{\leq} \dot{\cup} T^* \dot{\cup} \mathcal{T}_{\text{rev}}^* \dot{\cup} \mathcal{L}^* \dot{\cup} \mathcal{D}^* \dot{\cup} \mathcal{E}^* \dot{\cup} \mathcal{B}_{\text{rel}}^*$$

is sufficient: every operator in $\mathcal{A}_P$ relevant to MR derivation is assigned to at least one block. Hypothesis ?? is an empirical hypothesis open to refutation.

REMARK 1 (SINGLE-CLASS INSTANCES). For each of the three operator algebras instantiated in this paper ($\mathcal{A}_{\text{Boltz}}$, $\mathcal{A}_{\text{equi}}$, $\mathcal{A}_{\text{rel}}$), we adopt a canonical π -template per non-empty block under which $|\mathcal{I}_s / \sim_s| = 1$ (Boltzmann’s G block, for example, is treated as a single $\sim_s$ class with a π -template covering both geometric quarter-rotations and energy-group permutations). Under this instance-level grouping, $|\mathcal{M}(\mathcal{A}_P)|$ equals the number of non-empty blocks for each instance, so the empirical material that reports a fixed per-instance MetaPattern count (§§??, ??, ??) refers to these per-instance class counts and not to a general single-class assumption. A finer treatment (separating geometric and energy-group symmetry into two $\sim_s$ classes, for example) is permitted by Step 3 of CONSTRUCT-MP and would split the corresponding MetaPattern accordingly.

REMARK 2 (BLOCK SUFFICIENCY VS. Translate SUFFICIENCY). Hypothesis ?? asserts that the eight blocks suffice to assign every operator in $\mathcal{A}_P$ relevant to MR derivation. It does not assert that every MR formulable over $\mathcal{A}_P$ ’s operators is reachable through the framework’s Translate operator from a single block invariant. Section ?? exhibits, on the PWR core diffusion algebra $\mathcal{A}_{\text{PWR}}$, two MRs whose constituent operators are individually assigned to blocks of $\mathcal{D}(\mathcal{A}_{\text{PWR}})$ (rod-insertion semigroup; boration and moderator-temperature scaling), yet whose MR content is not in $\text{MR}(\mathcal{A}_{\text{PWR}})$ in the sense of Definition ?. The block decomposition is therefore a necessary but not sufficient input to MetaPattern construction: Translate’s expressive form (single-block invariants, first-order π -template, single partial-order direction, operating on $P(x)$ tuples rather than on operator-spectrum quantities) is a second, independent constraint on the framework’s reach. This distinction between “block sufficiency” (Hypothesis ??) and “Translate sufficiency” (open) is made explicit in §??.

REMARK 3 (KNOWN AND CONJECTURED OUT-OF-SCOPE PROGRAM FAMILIES). The hypothesis is open to refutation. Six families are known or conjectured to fall outside its image and to require an additional block:

- (1) Symplectic systems. Hamiltonian dynamics whose volume-preserving structure is captured by a symplectic 2-form rather than self-adjointness (N -body simulators, celestial-mechanics integrators).
- (2) Sheaf-theoretic / categorical constructions. Programs whose semantics are captured by functors and natural transformations between categories (certified compilers, type-driven program transformations); MRs of the form “commutativity of a square in the underlying category” do not reduce to single-block invariants under the present Translate.
- (3) Probabilistic / martingale invariants. MRs of the form “ $\mathbb{E}[f(\mathbf{x}_t) \mid \mathcal{F}_s] = f(\mathbf{x}_s)$ on a stopping-time domain” arise in MCMC samplers and stochastic gradient analyses.
- (4) Topological invariants. MRs of the form “ f ’s level sets have a fixed Betti-number signature” (topological data analysis, shape classifiers) require homological structure absent from the present blocks.
- (5) Label-consistency for supervised learning. MRs that constrain a model’s predictions against ground-truth labels (e.g. wrong-sign-loss detection on classifiers); the absence of a label-consistency operator is one reason the §?? case study reports zero detection on category-(i) wrong-sign mutations.
- (6) Empirical parameter-distribution divergence. A probability-distribution divergence operator on parameter-space measures, motivated by the §?? pilot’s category-v-02/04/05 (activation change, bias removal, weight re-init) which

the eight blocks do not detect. An MR of the form $D_{KL}(p_\theta \| p_{\theta+\Delta\theta}) \leq \tau$ for $\|\Delta\theta\| \leq \epsilon$ would lie in such a block. Empirical witness: §??’s cat-v-02/04/05 mutations are not detected by any MR set including Set N; this is that pilot’s principal negative finding for Hypothesis ??.

Each family signals a candidate ninth block. The framework’s intended response is to admit the additional block when the program family demands it, rather than to extend $\mathcal{B}_{rel}^$ ’s scope or to absorb the case as “out of scope for MR testing” in some over-broad sense. Concrete examples of MRs that lie outside Theorem ??’s scope are catalogued in Appendix ??; one candidate ninth block (metric-stability M_{lip}) has an explicit Translate-template design (Remark ?? below).*

REMARK 4 (DOMAIN-LEVEL OUT-OF-SCOPE). *Beyond the candidate-ninth-block families of Remark ?? (which signal extensions within the framework’s general programme), the following program-family classes are out of scope for NOETHER in the present form because they admit no operator-algebraic representation at all:*

- General web applications (HTTP request/response programs, business-rule engines, content-management systems) whose semantics are state-machine or rule-driven rather than equational.
- RLHF reward models and large-language-model agents whose relevant invariants are statistical/measure-theoretic rather than algebraic (calibration, refusal consistency, sycophancy).
- Distributed-consensus protocols (Paxos / Raft / Byzantine variants) whose semantics are CRDT- or message-trace-based; the relevant invariants are linearisability and convergence rather than operator-algebraic.
- Compiler-internal optimisations (loop transformations, register allocation, instruction scheduling) whose correctness is expressed as a refinement relation on an operational semantics rather than as a property over a fixed input/output algebra.

These classes are not candidate ninth blocks; they are domains where the framework’s scope precondition (program family admits an explicit operator-algebraic description) is structurally absent. NOETHER’s reach is therefore consistent with what its scope precondition explicitly requires: SUTs from textbook-codified mathematical, physical, or relational-algebra domains are within reach, and SUTs from general software-engineering domains without such codification are not.

REMARK 5 (METRIC-STABILITY CANDIDATE NINTH BLOCK, M_{lip}). *Among the candidate ninth blocks of Remark ??, the metric-stability block has the most concrete Translate-template proposal. Define $M_{lip} = (\mathcal{X}, d_X) \rightarrow (\mathcal{Y}, d_Y)$ as the family of K -Lipschitz maps between two metric spaces; the induced MetaPattern m_{lip} is the equivalence class of pointwise stability inequalities $d_Y(P(x'), P(x)) \leq K \cdot d_X(x', x)$. The corresponding Translate template constructs a follow-up by metric perturbation $x' = x + \epsilon u$ ($\|u\| = 1, |\epsilon| < \delta$) and predicates a K -Lipschitz output bound. Canonical-block ordering would place M_{lip} after $\mathcal{B}_{rel}^*$ since metric structure is independent of the eight existing block invariants (no group action on the perturbation set, no partial order on inputs, no self-adjoint operator, no parametric refinement family, no idempotent-semiring rewriting). Theorem ??’s closure proof transfers without modification because m_{lip} is single-block algebraically derived. We do not commit to M_{lip} as part of the canonical decomposition in this paper; we record it in Appendix ?? as the most concrete sub-instance of Remark ?? item (iv) (topological invariants) that admits an immediate Translate template, and as the empirical orphan in the audit of §??.*

3.2 Deriving MR classes with CONSTRUCT-MP

3.2.1 *Algebra-induced metamorphic relations.* Before defining algebra-induced MRs we make explicit the Translate operator that converts a single block invariant into an executable MR. Without this definition, the closure result of Theorem ?? would quantify over an undefined construct.

DEFINITION 11 (BLOCK INVARIANT). Let $s \in \mathcal{D}(\mathcal{A}_P)$ be one of the eight blocks. A block invariant of $\mathcal{A}_P$ under s is a pair $\iota = (\Phi, \pi)$ where Φ is a finite set of operators drawn from s , and π is a relation $\pi \subseteq (\mathcal{X} \times \mathcal{Y})^k$ for some arity $k \geq 1$ such that for every $P \in \mathcal{F}$ and every choice of operators $\phi_1, \dots, \phi_n \in \Phi$, the tuple $((x_i, P(x_i)))_{i=1}^k$ obtained by applying $\phi_1, \dots, \phi_n$ to a base input $x \in \mathcal{X}$ in the canonical order specified by s satisfies π . We write $\mathcal{I}_s$ for the set of block invariants under s and $\sim_s$ for the equivalence relation $\iota \sim_s \iota'$ iff $\Phi = \Phi'$ and π, π' define the same constraint up to relabelling of input/output coordinates.

The relation $\sim_s$ is reflexive, symmetric, and transitive by construction (set equality and constraint equality are equivalence relations); hence $\sim_s$ is a genuine equivalence relation on $\mathcal{I}_s$, justifying the quotient in Step 3 of CONSTRUCT-MP below.

DEFINITION 12 (Translate). The Translate operator is the function

$$\text{Translate} : \mathcal{I}_s \times \{s\} \longrightarrow \text{MR}(P)$$

that maps a block invariant $\iota = (\Phi, \pi)$ under block s to the metamorphic relation

$$\rho_{\iota, s} : \quad \forall x \in \mathcal{X}, \forall \phi_1, \dots, \phi_n \in \Phi : \quad \pi((x_i, P(x_i)))_{i=1}^k \text{ holds,}$$

where the input tuple $(x_i)_{i=1}^k$ is generated by applying the ϕ_j in the canonical order specified by s (the convention is fixed per block: e.g., for $s = G$, $x_i = g_i \cdot x_0$ with g_i enumerated by group orbit; for $s = T^*$, the tuple ranges over the two arguments of the inner product). Translate is well-defined because (i) the per-block canonical order resolves any ambiguity in tuple generation, and (ii) $\sim_s$ -equivalent invariants are mapped to MRs that are $\sim$ -equivalent as logical statements over P . Per-block instantiations of Translate are tabulated in Appendix ??.

DEFINITION 13 (ALGEBRA-INDUCED MR). Let $\mathcal{A}_P$ be a program-induced operator algebra and let ρ be an MR over a program P . We say ρ is induced by $\mathcal{A}_P$, written $\rho \in \text{MR}(\mathcal{A}_P)$, when there exist (i) a block $s \in \mathcal{D}(\mathcal{A}_P)$, (ii) a block invariant $\iota \in \mathcal{I}_s$, and (iii) the equality $\rho = \text{Translate}(\iota, s)$ holds in the sense of Definition ?. Properties of P that constrain executions but cannot be obtained as $\text{Translate}(\iota, s)$ for any single block s and any invariant $\iota \in \mathcal{I}_s$ are not algebra-induced in this sense; concrete examples are catalogued in Appendix ??.

3.2.2 Construction of the MetaPattern set. We present the deductive procedure CONSTRUCT-MP that maps an algebra-decomposition to a MetaPattern set.

Step 1: Invariant extraction. For each block s in $\mathcal{D}(\mathcal{A}_P)$, compute the set of invariants $\mathcal{I}_s$ of $\mathcal{A}_P$ under s .

Step 2: MR derivation. For each invariant $\iota \in \mathcal{I}_s$, derive the MR family $\mathcal{R}(\iota) = \{\rho \in \text{MR}(\mathcal{A}_P) \mid \rho = \text{Translate}(\iota', s), \iota' \sim_s \iota\}$.

Step 3: Quotient (per equivalence class). For each $\sim_s$ -equivalence class $[\iota] \in \mathcal{I}_s / \sim_s$, form the MetaPattern $m_{s, [\iota]} = \mathcal{R}(\iota)$.

Step 4: Aggregation. Return $\mathbb{M}(\mathcal{A}_P) = \bigcup_{s \in \mathcal{D}(\mathcal{A}_P)} \{m_{s, [\iota]} : [\iota] \in \mathcal{I}_s / \sim_s\}$, so $|\mathbb{M}(\mathcal{A}_P)| = \sum_s |\mathcal{I}_s / \sim_s| \geq$ the number of non-empty blocks.

REMARK 6 (METAPATTERN GRANULARITY). Each MetaPattern $m_{s, [\iota]}$ corresponds to a single $\sim_s$ -equivalence class of invariants in block s . A block carrying $k_s > 1$ such classes (for instance geometric quarter-rotations and energy-group permutations within G on $\mathcal{A}_{\text{Boltz}}$, when treated as separate $\sim_s$ classes) contributes k_s distinct MetaPatterns. In the three instances of §??-?? we group invariants within a block under a single $\sim_s$ -equivalence class (a single canonical π -template

per non-empty block; see Remark ??), so $|\mathbb{M}(\mathcal{A}_P)|$ equals the number of non-empty blocks in those instances. The Invariance-Blindness analysis (§??) applies directly to each $m_{s,[i]}$ at its native per-class granularity.

3.2.3 Algebraic closure under Translate, the canonical-block ordering, and out-of-scope MRs.

DEFINITION 14 (CANONICAL-BLOCK ORDERING). We adopt the strict total order on blocks

$$G > O_{\leq} > T^* > \mathcal{T}_{\text{rev}}^* > \mathcal{L}^* > \mathcal{D}^* > \mathcal{E}^* > \mathcal{B}_{\text{rel}}^*.$$

An MR derivable through multiple blocks is assigned to the highest-priority block in this order. The placement of $\mathcal{B}_{\text{rel}}^*$ at the bottom reflects its semiring-rewriting nature, which sits algebraically downstream of the seven physical-mathematical blocks: rewriting equivalences typically depend on the program family’s input-perturbation, order, and method-comparison structure rather than the converse.

THEOREM 1 (ALGEBRAIC CLOSURE UNDER Translate). Let $\mathcal{A}_P$ be a program-induced operator algebra with decomposition $\mathcal{D}(\mathcal{A}_P)$ as in Section ??, and let $\mathbb{M}(\mathcal{A}_P) = \text{CONSTRUCT-MP}(\mathcal{D}(\mathcal{A}_P))$. Then for every $\rho \in \text{MR}(\mathcal{A}_P)$ in the sense of Definition ??, there exists a unique $m \in \mathbb{M}(\mathcal{A}_P)$ such that $\rho \in m$, where uniqueness is determined under the canonical-block ordering.

REMARK 7 (SCOPE OF THEOREM ??). Theorem ?? is a closure statement, not a completeness claim over arbitrary properties one might assert about P ’s executions. It quantifies over $\text{MR}(\mathcal{A}_P)$ as defined by Definition ??, namely the algebra-induced MRs reachable through the Translate operator from a single block invariant. Three concrete classes of MRs lie outside this scope and are not covered by the theorem; they are documented in Appendix ??:

- (1) Probabilistic MRs without operator-algebraic representation: e.g. MRs that constrain output distributions (such as “the entropy of $f(\mathbf{x})$ should not decrease under input augmentation A ”) when the augmentation is not expressible as an operator in $\mathcal{O}$.
- (2) Input-distribution MRs not expressible over $\mathcal{A}_P$ operators: e.g. adversarial-perturbation MRs “ $\|\delta\|_p \leq \epsilon \Rightarrow f(\mathbf{x} + \delta) = f(\mathbf{x})$ ” whose perturbation set is not a group action on $\mathcal{X}$.
- (3) Compositional MRs spanning multiple blocks under non-Translate derivations: MRs requiring simultaneous use of two block invariants in a way that is not the canonical-block-ordering reduction of Definition ??.

The strictly stronger statement that every MR formulable as a property over $\mathcal{A}_P$ ’s operators (without restricting to Translate-reachable derivations) is contained in some $m \in \mathbb{M}(\mathcal{A}_P)$ is identified as Theorem 1’ in Appendix ?? . Section ?? and Appendix ?? establish that this stronger statement is false on the PWR core diffusion algebra $\mathcal{A}_{\text{PWR}}$, by exhibiting two concrete counterexamples (ρ_{nonadd} , $\rho_{\text{MTC-bor}}$) whose obstructions identify five structurally independent extensions of Translate’s signature. The combined open problem (whether such an extended Translate preserves Theorem ??’s closure and Theorem ??’s polynomial-time constructibility) is the principal open question for follow-up work.

A sceptical reading might object that the by-construction status of Theorem ?? makes it near-tautological. We acknowledge that the closure result is by-construction within the explicit scope of Definition ?? (which fixes $\text{MR}(\mathcal{A}_P)$ as the Translate-image of $\mathcal{A}_P$). The substantive value lies in what the theorem then enables: Theorem ?? converts an empirical-adequacy claim (“our pattern grid covers $X\%$ of observed MRs”) into a structural-adequacy claim (“our pattern grid is exhaustive of the algebra-induced MR space, modulo the explicit out-of-scope classes of Remark ??”). Empirical adequacy frameworks built on inductive pattern grids do not guarantee algebraic closure even in this bounded sense; NOETHER guarantees it, and the obligation imposed on any user of the framework is then to verify that no

Translate-reachable MR is dropped by CONSTRUCT-MP, a verification that is mechanical once the algebraic input has been fixed.

The theorem still imposes a checkable obligation on the framework. If an MR is induced by an invariant in $\mathcal{D}(\mathcal{A}_P)$ but cannot be assigned by CONSTRUCT-MP, then at least one component (invariant extraction, Translate, structural equivalence, or the canonical ordering) is underspecified. Thus the result is not an empirical coverage claim; it is a well-formedness and closure guarantee for the downstream construction once the algebraic input has been fixed. The framework’s additional theoretical results complement Theorem ?? : Theorem ?? bounds the construction’s polynomial-time complexity under a finite generating set; Theorem 1’ (Conjecture ??), which makes the stronger claim of absolute completeness over arbitrary properties expressible in $\mathcal{A}_P$, is falsified on $\mathcal{A}_{\text{PWR}}$ (§??); the falsification identifies five structural obstructions in Translate’s signature (Appendix ??), with five further candidate dimensions surveyed on $\mathcal{A}_{\text{equi}}$ and $\mathcal{A}_{\text{rel}}$ (§??).

REMARK 8 (SCOPE OF THEOREM ?? FOR THE $\mathcal{B}_{\text{rel}}^*$ BLOCK). *The closure result extends to $\mathcal{B}_{\text{rel}}^*$ provided the rewriting rule set $\mathcal{R}_{\text{rel}}$ is finite. For the relational-algebra fragment of TPC-H-class queries the rule set is finite by the standard heuristic-rewrite normal forms (selection-pushdown, projection-pushdown, join-reordering, distinct-elimination) [? ?], and Theorem ?? therefore covers algebra-induced MRs of this fragment. Query languages with unbounded recursive rewriting (e.g. Datalog with non-terminating rule sets) lie outside this scope and are catalogued as out-of-scope under Remark ??.*

3.2.4 Complexity and decidability.

THEOREM 2 (COMPLEXITY OF CONSTRUCT-MP UNDER A FINITE GENERATING SET). *Suppose $\mathcal{A}_P$ admits a finite generating set $\text{gen}(\mathcal{A}_P)$ of cardinality n , with each generator’s invariant computation assumed to terminate in time t_i . Then $\mathcal{M}(\mathcal{A}_P)$ is computable in time $\mathcal{O}(n \cdot \max_i t_i \cdot \log n)$. This is a complexity bound on the construction rather than a decidability result: termination of per-block invariant extraction is assumed within t_i , and for the relational block $\mathcal{B}_{\text{rel}}^*$ general query equivalence is undecidable, so only the fixed-rule-set fragment is covered (Remark ??). The factors t_i absorb the realised per-generator invariant cost: for a finite symmetry group the relevant t_i can scale with the group order $|G|$, which may itself grow exponentially in the number of generators (for example $(\mathbb{Z}_2)^k$ has $|G| = 2^k$), so the bound is polynomial in n and the realised generator costs (output-sensitive) rather than necessarily polynomial in a compact group presentation.*

REMARK 9 (SCOPE OF THEOREM ?? FOR THE $\mathcal{B}_{\text{rel}}^*$ BLOCK). *The polynomial-time bound holds for $\mathcal{B}_{\text{rel}}^*$ when the rewriting rule set $\mathcal{R}_{\text{rel}}$ is finite. For first-order SQL with bag semantics, query equivalence is undecidable in general, but query equivalence under a fixed rule set $\mathcal{R}_{\text{rel}}$ is decidable: practical SMT-based solvers verify equivalence on substantial benchmark fractions in the published literature [? ?]. Theorem ?? therefore covers the rule-set-bounded fragment of relational algebra; outside this fragment, decidability is governed by the underlying first-order theory and is not the framework’s responsibility.*

Per-block invariant-extraction cost.

On infinite groups. The first row of Table ??, $\mathcal{O}(|G|^2)$, is the natural bound for finite symmetry groups and is the regime under which Theorem 2 was originally stated. For a Lie group G such as $\text{SO}(3)$, $|G|$ is uncountable and the bound is replaced by $\mathcal{O}(d_G^2)$, where $d_G = \dim_{\mathbb{R}} \mathfrak{g}$ is the real dimension of G ’s Lie algebra. In the equivariant-ML instantiation of Section ??, $d_{\text{SO}(3)} = 3$; the symmetry-block invariant is computed once over a basis of three infinitesimal generators, and orbit closure is obtained through finite-dimensional linear algebra. For finitely generated infinite discrete groups (e.g. $\mathbb{Z}$, the integer-translation group of an unbounded grid solver), CONSTRUCT-MP requires a separate truncation